

Advanced Econometrics

Multinomial Logit, Conditional Logit

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Categorical Dependent Variables

Many empirical models involve dependent variables that are **categorical** rather than continuous.

Examples: Satisfaction levels, Transportation choices, Health status

Two common models: **Ordered Logit** (Ordinal Logit), **Multinomial Logit** (Unordered Logit)

Key distinction:

- **Ordered Logit:** Categories have a natural ranking (e.g., low < medium < high)
- **Multinomial Logit:** Categories have no inherent order

Statistical test distinction:

- **Ordered Logit:** Requires **proportional odds assumption**
 - Tested using **Brant test**
- **Multinomial Logit:** Requires **IIA assumption**
 - Tested using **Hausman or Small-Hsiao tests**

What if assumptions fail?

- If proportional odds is violated:
 - → **Generalized / partial ordered logit** (preferred)
 - → **Multinomial logit** (flexible fallback)
- Trade-off: Multinomial logit ignores ordering and is less efficient

Ordered Logit Model

Used when outcome categories have a **natural ranking**.

Example: Satisfaction scale

- 1 = Very dissatisfied
- 2 = Dissatisfied
- 3 = Neutral
- 4 = Satisfied
- 5 = Very satisfied

Ordered logit assumes an underlying latent variable:

$$y^* = X\beta + \varepsilon$$

Observed outcomes arise from thresholds:

$$y = 1 \text{ if } y^* < \tau_1, \quad y = 2 \text{ if } \tau_1 < y^* < \tau_2, \quad \dots$$

The model estimates coefficients β and cut-points τ_j .

Multinomial (Unordered) Logit

Used when categories have **no natural order**.

Example: Transportation choice

- Car
- Bus
- Train
- Bicycle

The model compares each outcome to a **reference category**.

Example (Car as base):

$$\log \left(\frac{P(Bus)}{P(Car)} \right) = X\beta_{Bus}$$

$$\log \left(\frac{P(Train)}{P(Car)} \right) = X\beta_{Train}$$

Each category has its own coefficient vector.

Key Differences

Feature	Ordered Logit	Multinomial Logit
Category order	Yes	No
Example	Satisfaction scale	Transport choice
Parameters	One β + thresholds	Separate β 's
Efficiency	Higher if ordering is valid	More flexible

Proportional Odds Assumption

The ordered logit relies on the **proportional odds** (or **parallel lines**) assumption:

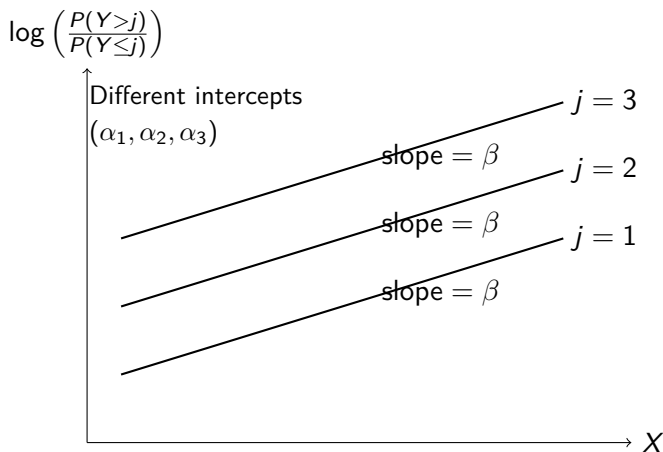
$$\log \left(\frac{P(Y > j)}{P(Y \leq j)} \right) = \alpha_j + X\beta$$

Key point:

- α_j changes across thresholds
- β is the **same for every threshold**

So each explanatory variable shifts the odds of being in a higher category by the **same amount**, regardless of where the cut-point is.

Visualization: Parallel Lines Assumption



- Different thresholds $\Rightarrow$ different intercepts
- Same effect of X across thresholds $\Rightarrow$ parallel lines

How to Read the Visualization

Suppose X is **income** and Y is **life satisfaction**.

Then ordered logit models:

- $P(Y > 1)$: more than very dissatisfied
- $P(Y > 2)$: more than dissatisfied
- $P(Y > 3)$: more than neutral

The figure shows:

- Each threshold has a different intercept
- But the effect of income is identical across thresholds
- Hence the lines are parallel

When the Assumption Fails

If the slopes are *not* the same across thresholds, the proportional odds assumption is violated.

Example:

- Income may strongly reduce the chance of being in the lowest category
- But have only a small effect on reaching the very top category

Possible alternatives:

- Generalized ordered logit
- Partial proportional odds model
- Multinomial logit

Multinomial Logit, Conditional Logit

People often choose **one option from several alternatives**.

Examples:

- Transport: Car / Bus / Train
- Lunch: Pizza / Burger / Salad
- Phone brand: Apple / Samsung / Xiaomi

Goal: model the **probability of choosing each option**.

Multinomial Logit (MNL)

Idea: choice depends on **characteristics of the person**.

Utility:

$$U_{ij} = X_i\beta_j + \varepsilon_{ij}$$

where:

- X_i = characteristics of person i
- β_j = parameters for alternative j
- ε_{ij} is typically assumed i.i.d. extreme value (Gumbel)

Choice probability:

$$P_{ij} = \frac{\exp(X_i\beta_j)}{\sum_{k=1}^J \exp(X_i\beta_k)}$$

Intuition for MNL

MNL asks:

How do person characteristics affect which option is chosen?

Examples of person variables:

- income
- age
- gender
- education

Example:

- richer students may choose salad more often
- younger students may choose burgers more often

Memory trick:

MNL $\rightarrow$ person characteristics

Conditional Logit (CL)

Idea: choice depends on **characteristics of the alternatives**.

Utility:

$$U_{ij} = Z_{ij}\beta + \varepsilon_{ij}$$

where:

- Z_{ij} = characteristics of alternative j for person i
- β = common parameter vector

Choice probability:

$$P_{ij} = \frac{\exp(Z_{ij}\beta)}{\sum_{k=1}^J \exp(Z_{ik}\beta)}$$

Intuition for Conditional Logit

Conditional Logit asks:

How do option characteristics affect the probability of being chosen?

Examples of alternative variables:

- price
- travel time
- distance
- comfort

Example: transport choice

- higher price $\Rightarrow$ lower probability
- longer travel time $\Rightarrow$ lower probability

Memory trick:

Conditional Logit $\rightarrow$ alternative characteristics

MNL vs Conditional Logit

Model	Main variables	Main question
Multinomial Logit	Person variables	Who chooses what?
Conditional Logit	Alternative variables	What option is attractive?

Examples:

- MNL: income, age, gender
- CL: price, time, distance

The IIA Problem

IIA = **I**ndependence of **I**rrelevant **A**lternatives

Definition:

The odds ratio between two alternatives does **not depend** on other available alternatives.

$$\frac{P_{ij}}{P_{ik}} = \frac{\exp(V_{ij})}{\exp(V_{ik})}$$

This ratio is unchanged when a new option is added or removed.

This is often unrealistic in real life.

Why IIA Can Be Unrealistic

Suppose a person chooses between:

- Car
- Bus

If a second bus that is almost identical appears, we expect:

- the two buses mostly steal probability from each other
- car probability should not collapse too much

But MNL treats alternatives in a way that often implies unrealistic substitution.

Classic Red Bus / Blue Bus Example

Original choices:

- Car
- Bus

Suppose:

- Car = 50%
- Bus = 50%

Now add a second identical bus:

- Car
- Red Bus
- Blue Bus

MNL prediction:

- Car = 33%
- Red Bus = 33%
- Blue Bus = 33%

This is the famous **IIA problem**.

Why Conditional Logit Improves on MNL

Conditional logit is often more useful because it includes **attributes of alternatives**.

Example:

- cost
- travel time
- comfort

So it can explain **why one option is preferred over another** in a more realistic way.

Important exam point:

Conditional logit improves MNL by using alternative-specific variables, but it does not fully remove IIA.

What Conditional Logit Fixes — and What It Does Not

Conditional Logit fixes:

- MNL cannot naturally use attributes like price or time
- CL directly models those alternative characteristics
- CL is better for transport, route, store, or product choice

Conditional Logit does **not** fix:

- strict IIA assumption still remains
- close substitutes can still be handled poorly

So CL is better than basic MNL for many applications, but still limited.

Mixed Logit (Random Parameters Logit)

Mixed Logit is a more flexible model.

Utility:

$$U_{ij} = X_{ij}\beta_i + \varepsilon_{ij}$$

where coefficients vary across individuals:

$$\beta_i \sim f(\beta)$$

Key idea:

- people have different tastes
- coefficients are random, not fixed for everyone

Intuition for Mixed Logit

Example: price sensitivity

- Some people strongly dislike expensive options
- Some people care less about price

So instead of one common coefficient for price, Mixed Logit allows:

$$\beta_{price,i}$$

to vary across people.

This captures **preference heterogeneity**.

Mixed Logit Choice Probability

Choice probability:

$$P_{ij} = \int \frac{\exp(X_{ij}\beta)}{\sum_{k=1}^J \exp(X_{ik}\beta)} f(\beta) d\beta$$

Interpretation:

- first compute logit probability for a given β
- then average over the distribution of β

This integral usually has no closed form, so estimation uses simulation.

Common method:

- maximum simulated likelihood

Why Mixed Logit Is Powerful

Mixed Logit solves important limitations:

- allows heterogeneous preferences
- relaxes the strict IIA assumption
- captures more realistic substitution patterns
- can allow correlation across alternatives

That is why Mixed Logit is one of the most flexible discrete choice models.

Model whcih I use in my research :)

Fertility timing model:

- We employed **mixed-effects complementary log-log (cloglog)** models
- Used for **transition to first and second birth**
- Appropriate for **discrete-time hazard models**
- Mixed-effects → captures **unobserved heterogeneity across individuals**

Why Mixed Logit is powerful (discrete choice context):

- Allows **heterogeneous preferences** across individuals
- Relaxes the strict **IIA assumption**
- Captures more realistic **substitution patterns**
- Allows **correlation across alternatives**

Key clarification:

- Mixed cloglog $\neq$ Mixed logit
- Cloglog is used for **timing/hazard models**
- Mixed logit is used for **choice among alternatives**

Nested Logit

Nested Logit is another way to deal with IIA.

Idea:

- group similar alternatives into **nests**
- IIA holds **within a nest**, but not necessarily across all alternatives

Example: transport choice

- Private transport: Car, Motorcycle
- Public transport: Bus, Train, Metro

This is more realistic when some options are naturally closer substitutes.

Why Nested Logit Helps

In MNL, all alternatives are treated too symmetrically.

Nested Logit allows:

- stronger substitution among similar choices
- weaker substitution among very different choices

Example:

- Bus and Train are closer substitutes
- Car may be less closely related to either of them

So Nested Logit partially relaxes the IIA problem.

Model Comparison

Model	Main variables	IIA?	Heterogeneity?
Multinomial Logit	Person	Yes	No
Conditional Logit	Alternative	Yes	No
Nested Logit	Alternative / grouped	Partial	Limited
Mixed Logit	Both possible	No (relaxed)	Yes

Summary

- **MNL**: choice depends on **who the person is**
- **Conditional Logit**: choice depends on **what the options are like**
- **Nested Logit**: similar options are put in **groups**
- **Mixed Logit**: different people can have **different tastes**

Exercise 1

We want to estimate a **multinomial logit model** for fishing mode choice.

The original dataset `Fishing` is stored in **wide format**, but the `mlogit` package requires data in **long format**.

Main question:

- Why does `mlogit.data()` change the structure of the dataset?
- What do the new variables such as `alt`, `chid`, and `mode` mean?

Original Code

```
Fish <- mlogit.data(Fishing, shape = "wide",  
                  choice = "mode", varying = 2:9)  
  
mlogit1 <- mlogit(mode ~ 0 | income, data = Fish)  
  
summary(mlogit1)
```

This code does two things:

- 1 Converts the data into the format needed by `mlogit`
- 2 Estimates a multinomial logit model using `income`

What the Original Data Looks Like: Wide Format

In the original dataset, **each row corresponds to one individual (or one choice situation)**.

Example:

mode	price.beach	price.boat	price.pier	price.charter	catch.beach	income
charter	157.930	157.930	157.930	182.930	0.0678	7083.332

Interpretation:

- The chosen mode is charter
- The prices for all alternatives are stored in separate columns
- The catches for all alternatives are also stored in separate columns

Why Wide Format Is Not Enough

A multinomial logit model compares **alternatives within each choice situation**.

The utility for individual i choosing alternative j is usually written as:

$$U_{ij} = \beta_1 \textit{price}_{ij} + \beta_2 \textit{catch}_{ij} + \beta_3 \textit{income}_i + \varepsilon_{ij}$$

Notice:

- $\textit{price}_{ij}$ and $\textit{catch}_{ij}$ vary across alternatives
- $\textit{income}_i$ is the same for all alternatives of the same person

So the model needs data indexed by:

(individual i , alternative j)

That is why we need **long format**.

What Long Format Means

After using `mlogit.data()`, each row becomes:

one individual $\times$ one alternative

For one person, instead of 1 row, we now get 4 rows:

- beach
- boat
- charter
- pier

So the data expands because every choice situation is rewritten as all available alternatives.

Output After Conversion

```
head(Fish[1:5,])
```

	mode	income	alt	price	catch	chid	idx
1	FALSE	7.083332	beach	157.930	0.0678	1	1:beach
2	FALSE	7.083332	boat	157.930	0.2601	1	1:boat
3	TRUE	7.083332	charter	182.930	0.5391	1	1:charter
4	FALSE	7.083332	pier	157.930	0.0503	1	1:pier
5	FALSE	1.250000	beach	15.114	0.1049	2	2:beach

Interpretation of the Long Data

For `chid = 1`, the dataset becomes:

chid	alt	price	catch	income	mode
1	beach	157.930	0.0678	7.083332	FALSE
1	boat	157.930	0.2601	7.083332	FALSE
1	charter	182.930	0.5391	7.083332	TRUE
1	pier	157.930	0.0503	7.083332	FALSE

This means:

- person 1 considered 4 alternatives
- charter was chosen
- all other alternatives were not chosen

Meaning of Each Variable in Fish

mode Logical variable:

- TRUE if the alternative was chosen
- FALSE otherwise

income Individual-specific variable:

- same value repeated for all alternatives of the same person

alt Name of the alternative:

- beach, boat, charter, pier

price Price of that specific alternative

catch Expected catch for that specific alternative

chid Choice situation ID

Why Does Income Repeat?

Look at the first four rows:

chid	alt	income
1	beach	7.083332
1	boat	7.083332
1	charter	7.083332
1	pier	7.083332

Income repeats because it is an **individual-specific variable**.

That means:

- it belongs to the person
- not to a particular alternative

So for the same person, income must appear in every row corresponding to that person's alternatives.

Comparing Wide and Long Format

Wide format

- 1 row per individual
- alternatives stored in different columns
- example:
 - `price.beach`
 - `price.boat`
 - `price.pier`
 - `price.charter`

Long format

- multiple rows per individual
- 1 row per alternative
- alternative identified by `alt`
- common variables repeated as needed

What Is the Role of mode?

In the original data:

- mode is a categorical variable
- it records the chosen alternative for each person

Example:

```
mode = charter
```

After conversion to long format:

- mode becomes a logical indicator
- TRUE for the chosen alternative
- FALSE for the non-chosen alternatives

So one category in wide format becomes a binary indicator in long format.

What Is chid?

chid stands for **choice ID**.

It identifies which rows belong to the same decision maker or the same choice occasion.

Example:

chid	meaning
1	first individual's choice situation
2	second individual's choice situation
3	third individual's choice situation

All rows with the same chid are grouped together when estimating the model.

What Is idx?

The `idx` column is an index combining:

- the choice situation
- the alternative

Example:

`1:charter`

This means:

- choice situation 1
- alternative = charter

It is mainly a technical indexing variable created by `mlogit`.

The Model Being Estimated

```
mlogit1 <- mlogit(mode ~ 0 | income, data = Fish)
```

Interpretation of the formula:

`mode    0 | income`

- Left side: `mode` is the dependent variable
- Before the bar: alternative-specific regressors
- After the bar: individual-specific regressors

So here:

- `0` means no alternative-specific explanatory variables are included
- `income` is the only explanatory variable

Why Is This Called a Pure Multinomial Model?

This model is called a **pure multinomial model** because it only includes **individual-specific variables**.

Here, the explanatory variable is:

income

There are no alternative-specific regressors such as:

- price
- catch
- travel cost
- time

So the model explains how an individual's income affects the probability of choosing each fishing mode.

Important Interpretation Point

Because `income` does not vary across alternatives for the same person, its coefficient cannot enter utility in exactly the same way for all alternatives.

Instead, `mlogit` estimates coefficients relative to a **base alternative**.

This means:

- the effect of `income` is measured relative to the omitted reference category
- coefficients show how `income` changes preference for one mode compared with the base mode

Data and Research Question

We observe **1822 purchases** of 2-liter soda bottles across **104 weeks** and **5 stores**. Consumers choose among:

- Pepsi (34.6%)
- 7-Up (37.4%)
- Coke Classic (28.0%)

Observed alternative-specific variables:

- **price**
- **feature** (promotion in flyer)
- **display** (special store display)

Goal: explain soda choice using a **conditional logit model**.

Why Not Multinomial Logit?

A standard multinomial logit model is not appropriate here because the regressors are **alternative-specific**.

In this dataset:

- Coke, Pepsi, and 7-Up can have different prices
- display and feature can differ by brand in the same shopping occasion

Therefore, utility depends on the **attributes of each alternative**, not only on consumer characteristics.

So we use the **Conditional Logit Model (McFadden)**:

$$U_{ij} = \beta_1 \text{price}_{ij} + \beta_2 \text{feat}_{ij} + \beta_3 \text{disp}_{ij} + \varepsilon_{ij}$$

$$P_{ij} = \frac{\exp(\beta X_{ij})}{\sum_{k=1}^J \exp(\beta X_{ik})}$$

R Model Specification

```
library(mlogit)
```

```
mlogit1 <- mlogit(soda ~ feat + disp + price | 0, data = cola2)  
summary(mlogit1)
```

Interpretation of the formula:

- Variables before | are **alternative-specific**
- Variables after | would be **individual-specific**
- | 0 means there are no individual-specific regressors

Estimated Conditional Logit Model

Variable	Estimate	Std. Error	p-value
feat	-0.0106	0.0799	0.8945
disp	0.4624	0.0930	< 0.001
price	-1.7445	0.1799	< 0.001

Interpretation:

- **price**: significant negative effect
- **display**: significant positive effect
- **feature**: not statistically significant

$$\text{Log-Likelihood} = -1822.2$$

Economic Interpretation of Coefficients

- A higher **price** lowers the probability that a soda is chosen.
- A **store display** raises the probability that the promoted soda is purchased.
- **Feature advertising** in the flyer has almost no effect in this sample.

This is consistent with demand theory:

- consumers are **price sensitive**
- promotion at the point of sale matters
- consumers substitute toward competing brands when one brand becomes less attractive

Marginal Effects for Price

For price, the estimated marginal effects matrix is:

	coke	pepsi	sevenup
coke	-0.3782	0.1711	0.2072
pepsi	0.1711	-0.3721	0.2011
sevenup	0.2072	0.2011	-0.4083

How to read rows:

- Each row shows the effect of a **\$1 increase in one soda's price**
- Diagonal elements are **own-price effects** (negative)
- Off-diagonal elements are **cross-price effects** (positive substitution)

Interpretation of Price Marginal Effects

Example: row **sevenup**

- A **\$1 increase in SevenUp price** decreases the probability of choosing SevenUp by 0.4083, i.e. about **40.83 percentage points**
- At the same time, the probability of choosing Coke rises by 0.2072
- The probability of choosing Pepsi rises by 0.2011

This is a **large** effect because the baseline share of SevenUp is only 0.3743 (37.43%).

Thus, a \$1 increase is economically large relative to the average SevenUp price (\$1.12), so consumers strongly substitute toward competing sodas.

Marginal Effects for Display

Display marginal effects:

	coke	pepsi	sevenup
coke	0.1003	-0.0453	-0.0549
pepsi	-0.0453	0.0987	-0.0533
sevenup	-0.0549	-0.0533	0.1082

Interpretation:

- If Coke gets a display, its choice probability rises by about **10 percentage points**
- Competing brands lose probability
- Similar effects hold for Pepsi and SevenUp

Conclusion: display promotions significantly shift demand toward the promoted brand.

Percentage Points vs Percent

A percentage point change can represent either a small or a huge shift, depending entirely on the context and the base value being measured.

Key distinction:

- **Percentage points (pp)** = absolute difference between percentages
- **Percent (%)** = relative change compared with the original value

Example:

- From 10% to 12% = **2 percentage points**
- But also a **20% increase**

This is why marginal effects in choice models are often described in **percentage points**, not percent.

Why Percentage Point Changes Can Be Small or Huge

A percentage point change can be:

- **small**, if the original value is already large
- **huge**, if the baseline is small or the event is rare

Examples:

- A market share drop from 90% to 89%: only a 1 percentage point decline, often small in context
- A disease risk rise from 0.1% to 1.1%: a 1 percentage point increase, but a **1000% relative increase**

So when interpreting marginal effects like 0.4083, context matters:

- compare the effect to the **baseline probability**
- compare the variable change to its typical level

Key Takeaways

- The soda-choice problem requires a **conditional logit model**, not standard multinomial logit
- **Price** has a strong negative effect on demand
- **Display** has a strong positive effect
- **Feature** is not statistically significant
- Price marginal effects show clear **substitution across brands**
- Display marginal effects show that promotions **shift market shares**
- Always distinguish **percentage points** from **percent changes**
- A percentage point change may be small or huge depending on the **baseline**